



Session 5.2: **Off-Site Solutions Deep Dive** *Vietnam's DPPA* *Pricing Considerations*



About the Speaker

Cong Nguyen
Allotrope Partners
Vietnam Clean Energy Manager

Cong is a renewable energy professional with over a decade of experience. Prior to joining Allotrope, Cong cultivated extensive clean energy technical expertise by supporting the development of solar and wind farms, battery energy storage systems, and asset management projects. In his role with Allotrope, Cong supports corporate clients to evaluate and implement clean energy and decarbonization solutions across a range of technologies. This includes leading techno-economic assessments to determine costs and benefits for specific sites, managing projects and building stakeholder relationships, and supporting business development efforts in Vietnam. Cong holds a Bachelor's degree in Electromechanics Maintenance Engineering from Hung Yen University of Technology and Education and has advanced certifications in renewable energy and reliability engineering.

Session Roadmap

Follow the Money: From your EVN Bill to a Bankable DPPA

#	Module
1	The Baseline: What You Pay EVN Today
2	Virtual DPPA Money Flow: The Five-Line Bill
3	CfD Mechanics: What Drives Your Quoted Price
4	Developer Economics and the Three Gates
5	Case Studies 5 & 6
6	Key Decision Points and Wrap-up

By the end of this session you will be able:

1. Breakdown a virtual DPPA monthly settlement into its five components and compare it to your EVN bill
2. Explain how the CfD strike, market price (FMP), loss factors, and grid fees set your net cost
3. See why buyers, sellers and lenders each have a “gate” and why a deal must pass all three
4. Leave with realistic pricing expectations grounded in a real factory model

Please note session glossary on your table for technical terms and definitions

Scope: Virtual DPPA (grid CfD) only.



Module 1:

*The Baseline: What You Pay
EVN Today*

What You Pay EVN Today: TOU Is The Reference Price

Period	Hours	Versus Base
Off-peak	00:00 - 06:00	~0.52x
Normal	06:00 - 18:00	1.0x
Peak	18:00 - 23:00	~1.78x

Sunday rule: no peak period on Sundays (all hours normal).

Voltage	Off-peak	Normal	Peak
>=110kV	0.034	0.065	0.116
22-110kV	0.037	0.070	0.126
6-22kV	0.042	0.079	0.141
<6kV	0.049	0.092	0.164

(USD/kWh approx. 2025)

K_{pp} values: 1.0085 for 110 kV, and 1.0273 for 22-110 kV.

Why this matters for DPPA

- Every DPPA offer is judged against this bill: "business-as-usual" (BAU)
- Peak hours cost 3.4x off-peak: your **load shape** decides what "**average**" means for you
- **Residual energy** (load not covered by the DPPA) keeps paying these TOU rates

The savings benchmark

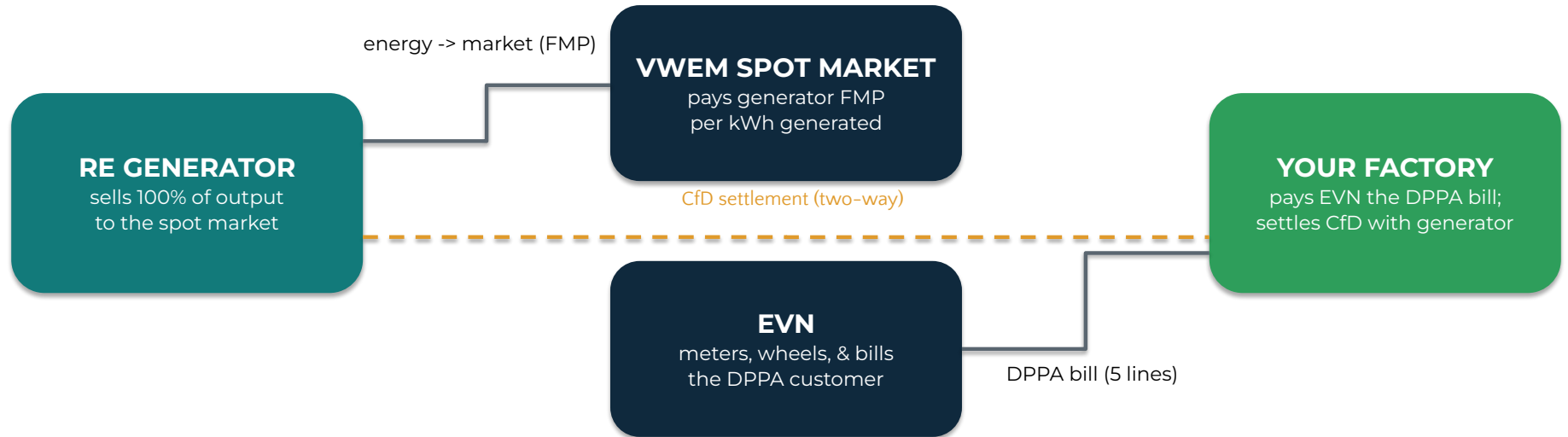
- BAU = your current EVN bill escalated **~4%/yr** (historical trend 2015-2024)
- A DPPA is evaluated on **cumulative cost vs BAU** over 10 years and over the **project lifetime** - not just Year 1
- EVN tariff escalation is the main source of long-term DPPA value



Module 2:

Virtual DPPA Money Flow: The Five-Line Bill

Virtual DPPA Money Flow: Three Relationships, One Bill



- Physical electrons flow through the national grid as usual and the DPPA is a financial overlay
- The generator's revenue = spot sales at FMP + CfD settlement with your factory
- Your factory cost = the five-line EVN bill + the CfD settlement

The Monthly DPPA Settlement Has Five Lines

#	Line	Formula	Level (2025 reference)
1	Market energy	$Q_{Khc} \times FMP \times k \times K_{pp}$	FMP avg ~1,426.6 VND/kWh; $k \sim 1.026$; K_{pp} by voltage
2	DPPA service fee	$Q_{Khc} \times C_{DPPADV}$	360 VND/kWh
3	Balancing fee	$Q_{Khc} \times P_{CL}$	163.3 VND/kWh
4	Residual purchase	$(\text{Load} - Q_{Khc}) \times \text{EVN retail}$	TOU rates
5	CfD settlement	$(\text{Strike} - FMP) \times \min(Q_C, Q_{Khc})$	Two-way: can be + or -

- Lines 1-4 are billed by EVN. Line 5 settles bilaterally with the generator under CfD contract.
- Q_{Khc} = the renewable volume actually received and consumed (defined next slide).
- $CFMP = FMP \times k$: k converts the market price to the customer delivery point (price conversion, not a volume loss).

Contracted Volumes & Actual Electricity Consumption

Which volume? You settle only on what you actually consume

$$Q_{\text{adj}} = Q_{\text{meter}} / K_{\text{pp}} \quad \text{delivered volume at your meter}$$

$$Q_{\text{Khc}} = \min(\text{Load}, Q_{\text{adj}}) \quad \text{settled volume (lines 1-3)}$$

$$Q_{\text{CfD}} = \min(Q_{\text{C}}, Q_{\text{Khc}}) \quad \text{CfD settles on received volume}$$

- Grid losses are reflected through coordinated volume and price conversions: K_{pp} converts generator-meter output into buyer-side credited volume, while k and K_{pp} convert FMP into the corresponding buyer-side market-energy price.
- If the plant generates more than you consume in an hour, the excess earns the generator spot revenue only; it is NOT billed to you and does NOT earn CfD
- **Practical consequence:** Contracted volume (Q_{C}) above your real consumption buys nothing

Simulation: Industrial Park Buys 6,000 MWh/month

Parameter	Symbol	Value
Consumption = received RE volume	$Q_{KH} = Q_{KHC}$	6,000,000 kWh/mo
Contracted CfD volume	Q_C	6,000,000 kWh/mo
CfD strike price	P_C	1,300 VND/kWh
Market price (monthly avg)	FMP	1,200 VND/kWh
Price conversion factor	k	1.026
Loss factor (110 kV)	K_{pp}	1.008
DPPA service fee	C_{DPPADV}	360 VND/kWh
Balancing fee	P_{CL}	163.3 VND/kWh
Avg retail price (residual)	P1	2,204 VND/kWh

Scenario

- A 50 MW solar plant contracts its full output to one industrial customer
- Consumption exactly equals delivered RE volume - no residual EVN purchase this month
- Strike just 100 VND above market - watch what the fees do

Source: EAVCED public training

The Five Lines, Computed

Line	Calculation	VND/month
1. Market energy (C_{DN})	$6,000,000 \times 1,200 \times 1.026 \times 1.008$	7,446,297,600
2. Service fee (C_{DPPA})	$6,000,000 \times 360$	2,160,000,000
3. Balancing fee (C_{CL})	$6,000,000 \times 163.3$	979,800,000
4. Residual (C_{BL})	$0 \text{ kWh} \times 2,204$	0
EVN bill (C_{EVN})	lines 1 + 2 + 3 + 4	10,586,097,600
5. CfD settlement (C_{CFD})	$(1,300 - 1,200) \times 6,000,000$	600,000,000
TOTAL (C_{KH})	$C_{EVN} + C_{CFD}$	11,186,097,600

- Effective cost = $11,186,097,600 / 6,000,000 = \sim 1,864$ VND/kWh for fully renewable supply (versus $\sim 2,204$ VND avg retail in the example)
- Generator side: $6,048,000 \text{ kWh} \times 1,200 \text{ (spot)} + 600,000,000 \text{ (CfD)} = 7,857,600,000$ VND

Your Monthly Settlement Has Five Lines

Component (per matched kWh)	VND/kWh	USD/kWh (cents)
Market energy at 2025 avg FMP x $k \times K_{pp}$ (22-110kV)	~1,504	~6.0
DPPA service fee (C_{DPPADV})	360	1.4
Balancing fee (P_{CL})	163.3	0.7
Delivered cost before CfD	~2,027	~8.1

- The 523 VND/kWh of fixed fees applies to every matched kWh, regardless of your strike
- They are set by regulation, not negotiation
- Compare ~2,027 VND delivered (pre-CfD) with your own TOU mix - for many factories that is already near or above average grid cost

Virtual DPPA is rarely a Day-1 discount. Its value comes from price certainty versus EVN escalation, being able to cover 100% of annual electricity demand with contracted renewable generation, and utilizing **credible renewable claims**.



Module 3:

*CfD Mechanics: What Drives
Your Quoted Price*

The CfD Is Two-Way: Certainty, Not Automatic Savings

$$\text{CfD cash flow} = (\text{Strike} - \text{FMP}) \times Q_{\text{CfD}}$$

Market condition	Who pays whom
FMP below strike	You top up the generator
FMP above strike	Generator pays you back

- Net effect: the renewable energy cost is locked near the strike (+ fees), whatever the market does
- 2025 reference: FMP averaged ~1,427 VND/kWh with strong hourly volatility; the CfD removes that volatility from the budget
- A strike below expected average FMP would mean expected receipts, but no developer can finance that (Module 4)

Strike Escalation & The Year 1 Reality

Strike escalation (e.g., 4%/yr)

- Escalating strikes improve the seller's and lender's lifetime economics...
- ...but do nothing for the Year 1 cost and compound against BUYER later
- Negotiate the escalation index as hard as the starting strike (fixed VND vs CPI vs USD-linked)

What Year 1 usually looks like:

- In most realistic structures, Year 1 DPPA cost is at or above BAU
- Savings build as EVN tariffs escalate (~4%/yr) against BUYER locked strike

→ **Advice: Evaluate on 10-year and lifetime cumulative cost, not on Year 1 alone**

Internal Budget Question to Consider: Can we accept a modest Year 1 premium to cap 15~25 year exposure?

Contract Volume VS Your Load Shape

Solar generation peaks midday, but the factory runs its own shape. The overlap—hour by hour - is what matters most.

If you...	Then...	Cost consequence
Over-contract ($Q_c > \text{consumption}$)	CfD caps at consumed volume - excess earns nothing	You paid contract terms for volume you can't settle
Match contract to hourly overlap	Every contracted kWh settles	Efficient; this is the target
Under-contract ($Q_c < \text{overlap}$)	More residual purchased from EVN at TOU	Less price certainty, smaller renewable claim

- Ask every developer for an hourly (8760 hour) matching analysis against your meter data, not an annual percentage
- Batteries can shift the overlap window but add CAPEX the tariff must carry (see the case study)



Module 4:

Developer Economics and the Three Gates

Developer's Perspective on Economics

Revenue =

$$Q_{\text{meter}} \times \text{FMP} +$$

spot sales (volatile)

$$(\text{Strike} - \text{FMP}) \times Q_{\text{CfD}}$$

CfD with your factory (stabiliser)

Capital Structure	70% Debt / 30% Equity
Debt Term	~10 years
Debt Interest Rate (VND)	~8.5% per year
Debt Interest Rate (USD)	~5% per year (International)
Equity Target Return	12 - 15%+ IRR
CIT Incentives	4 years exempt + 9 years at half rate

- The CfD converts volatile spot revenue into a near-fixed price on matched volume
- Your strike IS the developer's bankable revenue line
- Everything the lender checks flows from it

A Deal Must Pass “Three Gates” to Proceed

THREE GATES

Gate	Who	Test	Strike pressure
1. Buyer	You	Cumulative cost $\leq$ BAU over 10 yr AND lifetime	Pushes strike DOWN
2. Seller	Developer	Equity IRR $\geq$ 12-15%	Pushes strike UP
3. Lender	Bank	Minimum DSCR $\geq$ ~1.20x every year	Pushes strike UP (hardest gate)

- $$\text{DSCR} = \frac{\text{Cash available for Debt service}}{\text{Debt Service}}, \text{ computed for every year of the loan}$$
- One CAPEX heavy year (e.g., a battery replacement) can sink the whole financing even if averages look fine
- The negotiation window is the strike range where all three gates “pass”; but it can be empty (see case study)

A strike below the developer's bankability floor does not mean a cheap deal; it means no project.

Key Lifecycle Items The Tariff Must Contain

Item	Typical assumption	Buyer Implication
PV degradation	~0.5%/yr output decline	Matched volume shrinks; lost kWh repurchased from EVN at TOU
Battery replacement	Large CAPEX every 8-10 years	Can highly impact DSCR in that replacement year
CIT holiday	4 yr exempt + 9 yr half rate (RE projects)	Improves developer economics
FX exposure	VND strike vs USD capex/debt	USD-linked terms shift FX the risks; price it consciously

- The above terms are key components of the tariff and are included even when they are absent from the term sheet
- These rarely appear in headline term sheets, but each one moves the bankable strike floor and therefore your price.



Module 5:

Case Studies 5 & 6

Same Factory - Now Procuring Offsite

On-Site Session: Cases 1-4 explored rooftop solar + BESS behind the meter.

Now: The same southern Vietnam manufacturing facility (22-110kV) signs a **virtual DPPA** with a dedicated solar + storage project.

- **Case 5: Solar + Large BESS**
- **Case 6: Solar + Minimum BESS**

Deal frame (both cases)	Value
DPPA type	Virtual (grid CfD) under Decree 57/2025
Strike price	2,000 VND/kWh, escalating 4%/yr
Tenor / analysis period	25 years
Financing	70% debt, 8.5% VND, 10-yr term
Method	Full 8760-hour settlement simulation (ND57 formulas, hourly FMP/CFMP series) + 25-year project cash-flow model with the three gates.

Case Study 5: Solar + Large BESS

BATTERY EATS THE DEAL

Metric	Result
Seller equity IRR	16.9%
Project IRR	13.5%
Developer NPV	\$1.52M
Minimum DSCR	1.14x (replacement year)
Payback	9.1 yr
Buyer vs BAU - Year 1	-8.7%
Buyer vs BAU - 10-yr cumulative	-8.9%
Buyer vs BAU - lifetime	-9.3%

- At strike 2,000 VND/kWh the buyer pays ~9% MORE than BAU on every horizon
- The oversized battery adds a ~\$1.2M replacement around year 11 that the tariff must recover
- Even so, lender comfort is thin (min DSCR 1.14x < 1.20x)

Lesson 1: An oversized battery cannot carry its own replacement cost through the tariff; everyone loses.

Case Study 6: Solar + Minimum BESS

CHALLENGING, YET BANKABLE

Metric	Result
Seller equity IRR	26.9%
Project IRR	18.2%
Developer NPV	\$2.54M
Minimum DSCR	1.50x (comfortably > 1.20x)
Payback	4.7 yr
Buyer vs BAU - all horizons	-14.4%

- Lean sizing: seller and lender are now comfortable - returns strong, DSCR 1.50x
- But at strike 2,000 VND/kWh the buyer is ~14% worse than BAU: the seller's headroom IS the negotiation space
- Therefore, sweep the strike down and see where the gates cross

Lesson 2: Right-sized the BESS makes the project financeable. Then, the entire negotiation is the strike price.

Sweeping The Strike On Case Study 6: Empty Window

56 scenarios analyzed: 12 strike prices ranging from 1,200-2,200 VND/kWh and 4 contract volumes ranging from 70-100%.

Gate results:

Strike level (VND/kWh)	Buyer gate	Seller gate	Lender gate ($\geq 1.20x$)	Balanced?
~2,000 (offer)	FAIL -14%	PASS	PASS 1.50x	No
~1,400	FAIL -1.4%	PASS 19%	PASS 1.19-1.5x	No
~1,300 x 70% vol	PASS +0.5% lifetime	PASS 17.9%	FAIL 1.14x	No
~1,200	PASS +2.9%	PASS	FAIL $<1.20x$	No

The buyer turns positive just as the lender drops out: 0 of 56 scenarios pass all three gates at current market prices and fee levels.

Pricing alone cannot balance this deal, BUT there are “levers” that can: lower leverage, debt sculpting, USD-denominated debt (~5%), fee relief, longer tenor.

What This Means And Expectations

- **Expect low single-digit lifetime savings at best.** The product you are buying is 10-25 year price certainty, meet up to 100% of annual electricity demand with contracted renewable generation, and utilize credible renewable claims, but all at a price premium.
- **Expect Year 1 to cost more than BAU.** Plan the internal approval around cumulative horizons.
- **Strike offers near 2,000 VND/kWh leave the buyer deeply negative.** Realistic buyer-positive territory starts near ~1,300 VND, which today fails the lender.
- **Deal feasibility is decided by financing structure as much as by price.** Ask developers HOW their project is financed, not just what they charge.



Module 6:

*Key Decision Points and
Wrap-up*

Five Questions That Change The Term Sheet

#	Decision	What to check
1	Strike level & escalation/indexation	Fixed VND vs escalating/deescalating vs USD linked; model these vs BAU
2	Contract volume vs load shape	8760 hourly matching vs meter data; CfD caps at consumed volume
3	Tenor & termination	Developer wants 15-25 yr; match to your lease / order book; exit terms
4	Voltage eligibility & K _{pp}	>=22kV only today; which K _{pp} applies decides your loss adjustment
5	Risk allocation	FMP volatility, curtailment, negative CfD months, change-in-law

Items 1-4 are quantifiable before negotiation; run them against load data first.

Three Core Reminders

- **Know the five-line bill:** 523 VND/kWh of fees and the loss factor are fixed; the strike price remains the big number open to negotiation.
- **Evaluated the deal on 10-year and lifetime cost vs BAU** - not on Year 1
- **Every deal must pass buyer, seller AND lender gates:** If the window is empty, the fix is structural, not just price

Next: Panel discussion on risk allocation and the reality of bank financing for Grid DPPA

Questions & Answers

QR to be updated